

Ex.no:5B	DATA VISUALIZATION USING GOOGLE PALYSTORE DATASET

AIM:

To visualize dataset features and distributions using line charts, bar charts, pie charts, histograms, scatter plots, and box plots using Matplotlib.

ALGORITHM:

1. Start the program and import Pandas and Matplotlib libraries.
2. Load the Google Play Store dataset into a DataFrame.
3. Display the dataset shape to verify the loaded data.
4. Extract the first 20 non-null app ratings.
5. Plot the app ratings using a line chart.
6. Count the apps in each category and select the top 10 categories.
7. Plot the top 10 categories using a bar chart.
8. Select the top 5 categories and represent them using a pie chart.
9. Plot the distribution of app ratings using a histogram.
10. Display the visualizations and stop the program.

PROGRAM AND OUTPUT:

```

In [4]: import os

        print(os.listdir(r'C:\Users\Devadharshini\google_play_data'))

['googleplaystore.csv']

In [5]: import pandas as pd

        df = pd.read_csv(r'C:\Users\Devadharshini\google_play_data\googleplaystore.csv')

        print("Dataset loaded successfully!")
        print(df.shape)

Dataset loaded successfully!
(10840, 13)

```

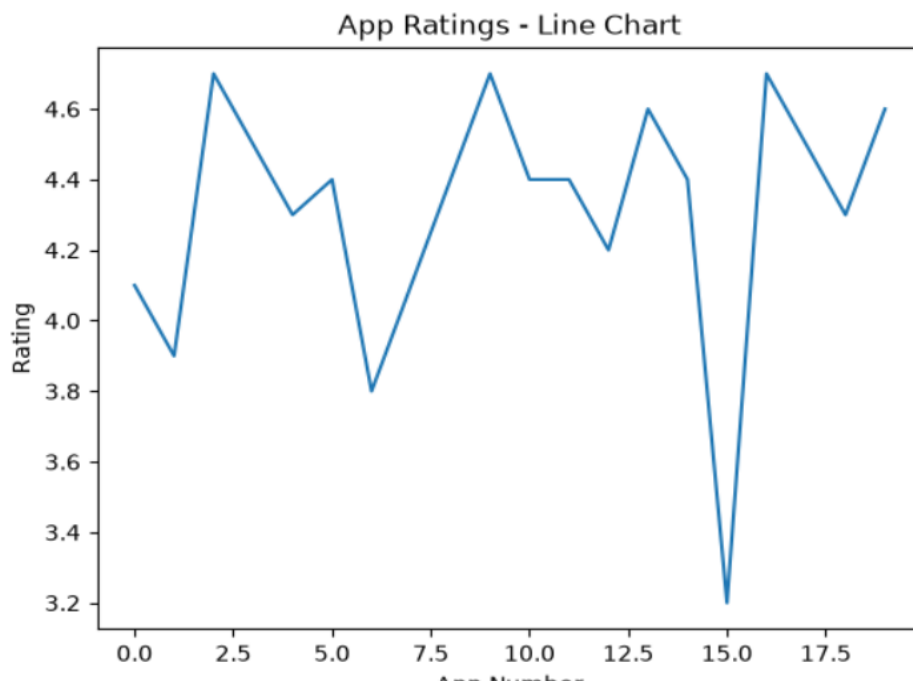
```
In [6]: import matplotlib.pyplot as plt

ratings = df['Rating'].dropna().head(20)

plt.plot(ratings)

plt.title("App Ratings - Line Chart")
plt.xlabel("App Number")
plt.ylabel("Rating")

plt.show()
```



```
In [7]: category_counts = df['Category'].value_counts().head(10)

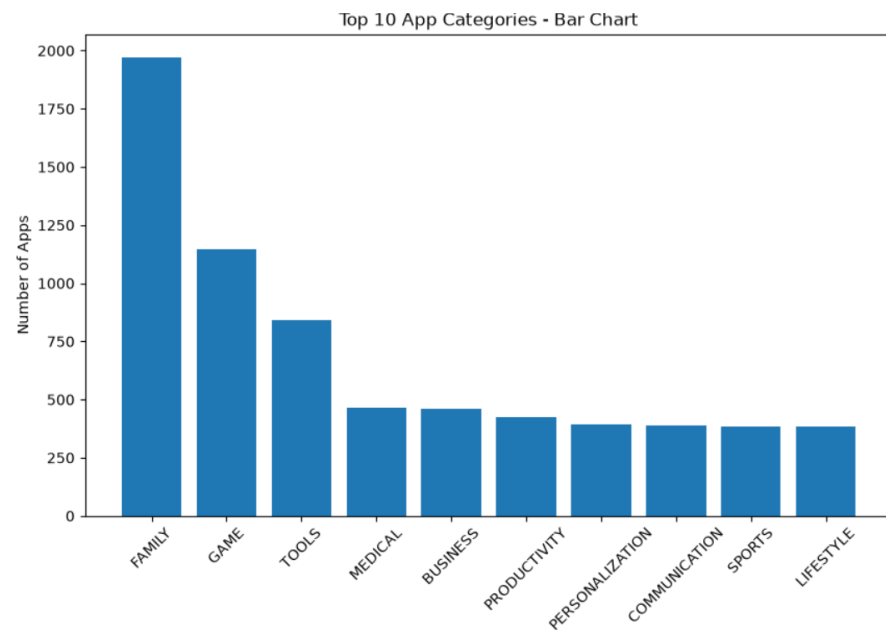
plt.figure(figsize=(10, 6))

plt.bar(category_counts.index, category_counts.values)

plt.title("Top 10 App Categories - Bar Chart")
plt.xlabel("Category")
plt.ylabel("Number of Apps")

plt.xticks(rotation=45)

plt.show()
```



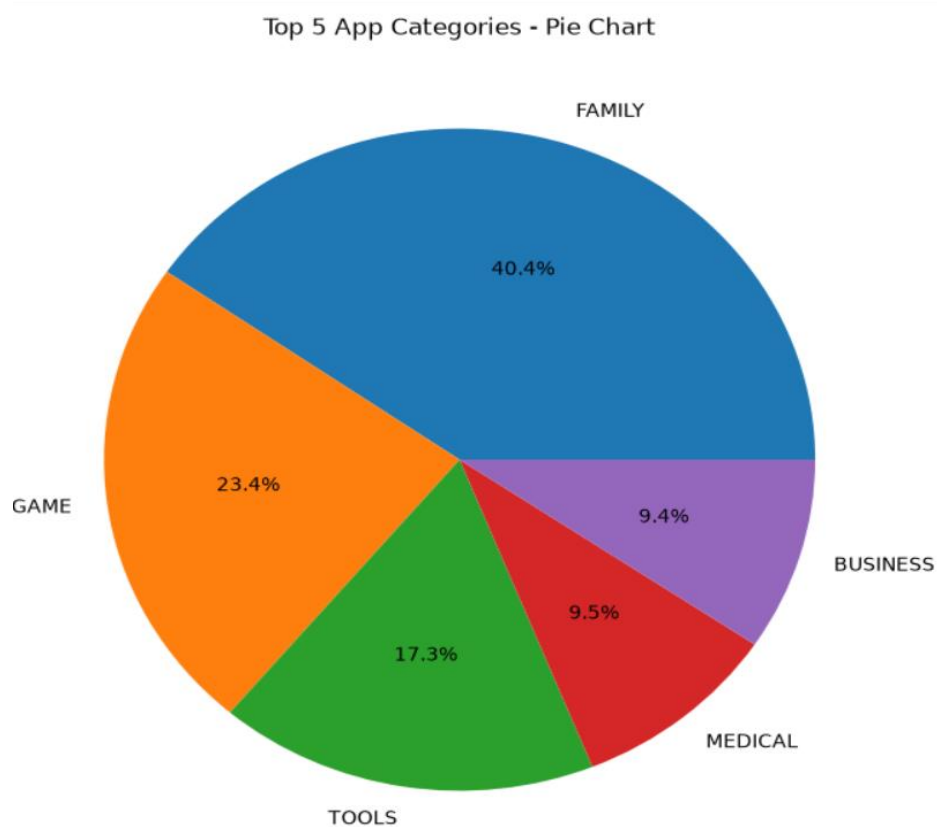
```
In [8]: category_counts = df['Category'].value_counts().head(5)

plt.figure(figsize=(8, 8))

plt.pie(
    category_counts.values,
    labels=category_counts.index,
    autopct='%1.1f%%'
)

plt.title("Top 5 App Categories - Pie Chart")

plt.show()
```



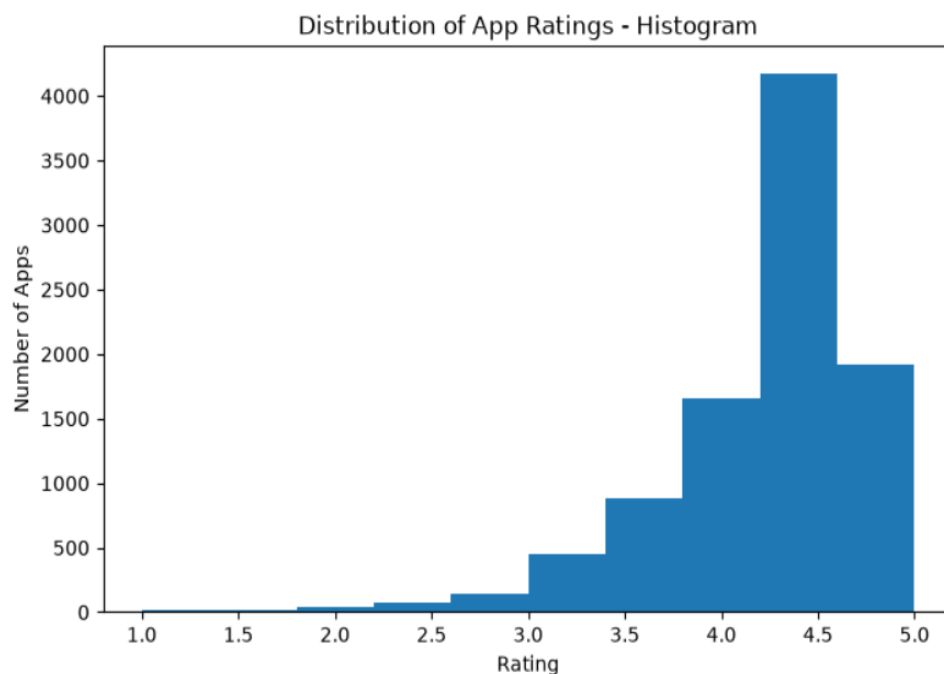
```
In [9]: ratings = df['Rating'].dropna()

plt.figure(figsize=(8, 5))

plt.hist(ratings, bins=10)

plt.title("Distribution of App Ratings - Histogram")
plt.xlabel("Rating")
plt.ylabel("Number of Apps")

plt.show()
```



```
In [10]: df['Installs'] = df['Installs'].str.replace('+', '', regex=False)
df['Installs'] = df['Installs'].str.replace(',', '', regex=False)
df['Installs'] = pd.to_numeric(df['Installs'], errors='coerce')

print(df[['Installs', 'Rating']].head())
```

	Installs	Rating
0	10000	4.1
1	500000	3.9
2	5000000	4.7
3	50000000	4.5
4	100000	4.3

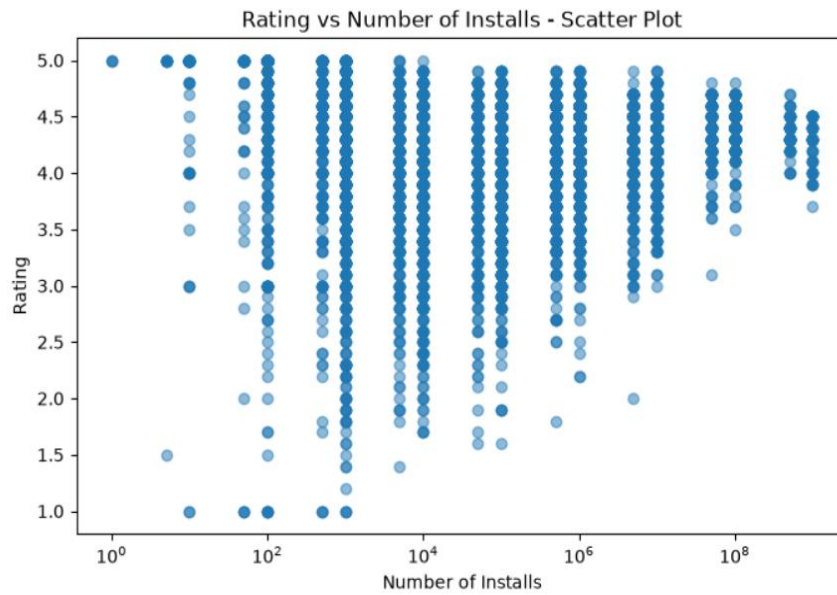
```
In [11]: plt.figure(figsize=(8, 5))

plt.scatter(df['Installs'], df['Rating'], alpha=0.5)

plt.title("Rating vs Number of Installs - Scatter Plot")
plt.xlabel("Number of Installs")
plt.ylabel("Rating")

plt.xscale('log')

plt.show()
```



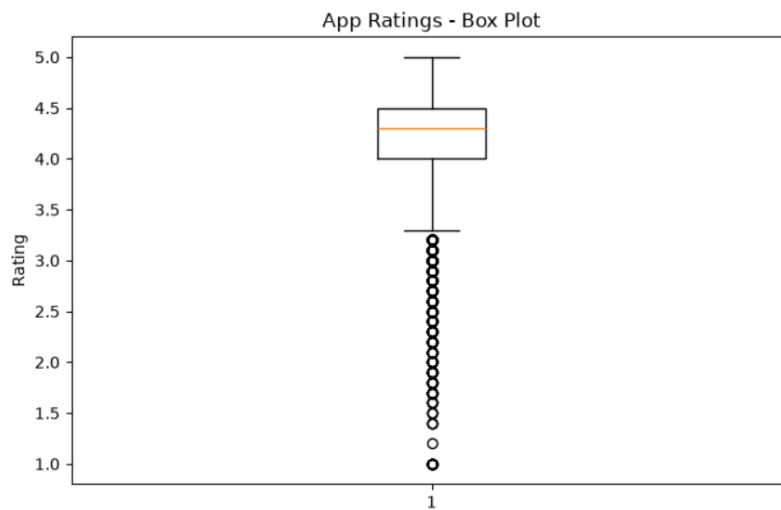
```
In [12]: ratings = df['Rating'].dropna()

plt.figure(figsize=(8, 5))

plt.boxplot(ratings)

plt.title("App Ratings - Box Plot")
plt.ylabel("Rating")

plt.show()
```



RESULT:

Thus dataset features and distributions using line charts, bar charts, pie charts, histograms, scatter plots, and box plots has been visualized successfully.

